

Joshua Paul Plan Guirao

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Education

Tufts University

September 2024-May 2028

B.S. in Computer Science & Quantitative Economics

Minors: Mathematics

Relevant Coursework: Intro Computer Science, Intermediate Microeconomic Theory, Data Structures, Calculus III, Discrete Mathematics, Web Programming, Linear Algebra, Probability, Foundations of Quantitative Economics, Intro to Machine Learning and Data Mining

Relevant Experience

Software Developer, JumboCode

Medford, MA, September 2025-Present

- Building a full-stack analytics dashboard for School On Wheels Massachusetts to automate reporting of social media, newsletter, and website performance, enabling data-driven decision-making and reducing manual reporting.
- Designed & implemented frontend components in React, Vite, and Tailwind CSS, creating responsive data visualizations for users.
- Developed backend services with Node.js, Express, PostgreSQL, & Prisma, enabling daily snapshot storage and trend analysis with APIs.
- Contributed to feature branch development, code reviews, and sprint planning to ensure the timely delivery of project milestones.

Researcher, Tufts Idea Lab

Medford, MA, September 2025-Present

- Contributing to Plastibot, a research initiative developing machine learning models with hyperspectral imaging to classify plastic types and create scalable, cost-efficient recycling solutions for real-world deployment.
- Collected and prepared hyperspectral image datasets by cutting and processing plastic samples to train machine learning classification models and analyze spectral data correlations.
- Refined Convolutional Neural Networks to classify six main plastic types using spectral band analysis, improving classification accuracy.

Data Science Intern, LeadUp Universe

Gurugram, New Delhi, June 2025-July 2025

- Spearheaded a strategic data initiative leveraging a 40,000-entry lead dataset to identify growth opportunities across geographic markets.
 - Designed and executed a full data pipeline, which involved cleaning and standardizing data using Python (pandas and NumPy) and Excel for analytical accuracy.
 - Conducted deep-dive analyses to uncover market trends and high-conversion customer segments, enabling data-driven market prioritization.
 - Delivered actionable insights through dashboards and visualizations (matplotlib), informing market strategies and regional targeting decisions.
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Projects

FitWise

Cambridge, MA, Oct 2025

- Developed FitWise, an AI-powered virtual fitness trainer that provides real-time form correction, rep counting, and motivational audio feedback using computer vision and AI at HackHarvard.
 - Implemented pose detection via Google's MediaPipe and integrated Gemini API for intelligent motion analysis to identify shallow reps and improper posture while using ElevenLabs' voice synthesis API to enable natural, interactive audio coaching, creating an authentic personal training experience.
 - Built a dual-process Python backend using Flask-SocketIO for seamless synchronization between pose detection and live updates, ensuring zero-lag real-time feedback.
 - Designed and deployed a Next.js frontend featuring a live skeleton overlay to visually guide users in maintaining proper form and track movement accuracy.
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Extra-Curricular Activities

Analyst, Tufts Trading Club

Medford, MA, September 2025-Present

- Study quantitative finance concepts such as probability, options, and pricing models to strengthen technical knowledge relevant to trading roles.
- Practicing algorithmic and probability-based problem solving to develop rapid analytical thinking and quantitative reasoning for technical interviews.

Chair Of Professional Development, Ascend Tufts

Medford, MA, November 2025-Present

- Organized professional development events, including skill-building and resume workshops, to support members' career growth.
- Coordinated outreach to industry professionals to host networking sessions and speaker events, enhancing club engagement and opportunities.
- Fostered member relations by gathering feedback and ensuring alignment of events with professional development goals.

Treasurer, Singaporean Indonesian Malaysian Student Association

Medford, MA, September 2024-Present

- Managed a \$7,000 annual budget, allocating funds efficiently across events and operations to ensure financial stability and maximize engagement.
 - Created and maintained budget forecasts and expense reports in Excel to improve accuracy and transparency of financial planning.
 - Worked with the leadership team to align financial decisions with the organization's mission and long-term objectives.
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Skills

Programming Languages

Python, Java, C++, SQL, HTML, CSS, JavaScript, VPython, Git, PHP, NoSQL, PostgreSQL

Software

Google Apps, Microsoft Suite, VS Code, Apache NetBeans, JupyterLab, GitHub, React, MongoDB

Languages

English (Fluent), Chinese (Intermediate), Filipino (Elementary)

Interests

Toronto Raptors, New England Patriots, Badminton, Piano, Weightlifting, Video Games, Chess, F1